

Supplementary Information

Sensitivity-aware inspection prioritisation for archaeological landscapes: a reproducible Taxila observatory

1 Supporting methods and complete experimental results

The main article presents the multitemporal maps, score baselines, ablations, support and joint-uncertainty experiments, climate comparison and seven-family land-cover analysis. This supplement provides full component inventories, numerical tables, configurations and secondary diagnostics. Spatial simulations and land-cover agreement have the distinct scopes specified in the main methods.

2 Component inventory and analytical support

Saraikala is preserved without invented geometry. Coordinates are inventory points, not property polygons. The same 17 mapped components are compared at all three radii.

Table S1: Complete Taxila component inventory and analytical geometry status.

ID	Component	Coordinate status	Longitude	Latitude	Included
139-001	Khanpur Cave	official_web_point	72.909294	33.799200	Yes
139-002	Saraikala, prehistoric mound	official_id_geometry_missing			No
139-003	Bhir Mound	official_web_point	72.819331	33.743944	Yes
139-004	Sirkap (fortified city)	official_web_point	72.829878	33.753403	Yes
139-005	Sirsukh (fortified ruined city)	official_web_point	72.849739	33.772256	Yes
139-006	Dharmarajika stupa and monastery	official_web_point	72.842667	33.744669	Yes
139-007	Khader Mohra (Akhuri)	official_web_point	72.860722	33.760806	Yes
139-008	Kalawan group of buildings	official_web_point	72.839753	33.746292	Yes
139-009	Giri complex of monuments	official_web_point	72.883325	33.729089	Yes
139-010	Kunala stupa and monastery	official_web_point	72.831444	33.750553	Yes
139-011	Jandial complex	official_web_point	72.828914	33.764456	Yes
139-012	Lalchak and Badalpur Buddhist stuppa	official_web_point	72.868019	33.781803	Yes
139-013	Mohra Moradu stupa and monastery	official_web_point	72.861053	33.761097	Yes
139-014	Pippala stupa and monastery	official_web_point	72.866686	33.765922	Yes
139-015	Jaulian stupa and monastery	official_web_point	72.875311	33.765042	Yes
139-016	Lalchak mounds	official_web_point	72.849469	33.774450	Yes
139-017	Buddhist remains around Bhallar stupa	official_web_point	72.825358	33.813497	Yes
139-018	Giri Mosque and tombs	official_web_point	72.880522	33.727769	Yes

Note: Coordinates are WGS 84 point locations transcribed from the official component map. Saraikala remains in the inventory but is excluded from coordinate-dependent analysis. Analytical circles are not official property polygons or legal buffers.

3 Annual climate analysis

The procedure resamples detrended residual blocks. A trend-restored ensemble supplies confidence intervals; a constant-baseline ensemble supplies null slopes. The 12 annual metrics form the multiplicity family. Heavy precipitation counts days exceeding the 1991–2020 wet-day 95th percentile (wet days have at least 1 mm rain); hot days exceed the baseline maximum-temperature 95th percentile. Heatwave duration is the longest run above that temperature threshold, without a separate minimum-duration criterion. Dry-spell duration counts consecutive days below 1 mm. Rx1day and Rx5day are the largest one-day and rolling five-day precipitation totals.

Table S2: Annual residual-block trends; 2,000 draws and five-year blocks.

Metric	Slope/year	CI low	CI high	p	Annual q
precipitation_sum_mm	-10.3037	-17.9150	-2.6679	0.0090	0.0270
rx1day_mm	-0.6233	-1.5426	0.4034	0.1549	0.2324
rx5day_mm	-1.7354	-3.3891	-0.2019	0.0395	0.0948
heavy_precipitation_days	-0.1429	-0.2266	-0.0582	0.0015	0.0180
consecutive_dry_days	0.1667	-0.1417	0.4322	0.2029	0.2597
temperature_mean_c	0.0418	0.0099	0.0731	0.0065	0.0260
annual_or_period_max_tmax_c	0.0333	-0.0079	0.0778	0.1204	0.2065
hot_days	0.1818	-0.1728	0.5227	0.3063	0.3342
heatwave_max_days	0.0323	-0.0449	0.1144	0.3493	0.3493
relative_humidity_mean_percent	-0.1311	-0.2852	0.0255	0.1124	0.2065
soil_moisture_mean_m3m3	-0.0013	-0.0022	-0.0003	0.0065	0.0260

Precipitation and evapotranspiration slopes are in mm year^{-1} , temperature slopes in $^{\circ}\text{C year}^{-1}$, humidity in percentage points year^{-1} , soil moisture in $\text{m}^3 \text{m}^{-3} \text{year}^{-1}$ and event counts in days year^{-1} . Effective-sample-size diagnostics use detrended lag-one correlation and are bounded by the observed record length; they do not replace the bootstrap.

4 Ranks and uncertainty scope

The following tables report component scores and two complementary sensitivity designs. The weight experiment perturbs the integrated domain architecture; the spatial experiment resamples 150 m spectral blocks with terrain fixed. They must not be described as the same probability distribution.

Table S3: Component scores and scale-dependent ranks.

Component	Rank 250 m	Rank 500 m	Rank 1,000 m	Score 500 m
139-001	15.0000	14.0000	11.0000	0.4309
139-003	14.0000	17.0000	17.0000	0.3397
139-004	10.0000	12.0000	15.0000	0.4799
139-005	8.0000	9.0000	10.0000	0.5015
139-006	3.0000	5.0000	6.0000	0.5363
139-007	7.0000	6.0000	3.0000	0.5353
139-008	12.0000	7.0000	9.0000	0.5191
139-009	4.0000	1.0000	2.0000	0.5912
139-010	5.0000	10.5000	13.0000	0.5010
139-011	16.0000	16.0000	16.0000	0.3485
139-012	17.0000	15.0000	12.0000	0.3922
139-013	11.0000	4.0000	5.0000	0.5392
139-014	6.0000	10.5000	8.0000	0.5010
139-015	2.0000	3.0000	1.0000	0.5676
139-016	13.0000	13.0000	14.0000	0.4461
139-017	9.0000	8.0000	7.0000	0.5172
139-018	1.0000	2.0000	4.0000	0.5725

Table S4: Spatial-block rank uncertainty; terrain held fixed.

Component	Median rank	Low	High	Top-3 frequency
139-017	1.0000	1.0000	5.0000	0.9110
139-009	2.0000	1.0000	3.0000	0.9855
139-018	3.0000	1.9750	5.0000	0.6150
139-015	4.0000	2.0000	8.5000	0.2940
139-006	6.0000	3.0000	11.0000	0.0385
139-013	8.0000	3.0000	15.0000	0.0510
139-014	8.0000	3.5000	12.0000	0.0245
139-010	8.0000	4.5000	12.0000	0.0055
139-007	9.0000	3.0000	15.0000	0.0465
139-008	9.0000	4.5000	14.0000	0.0040
139-005	9.0000	5.5000	13.0000	0.0005
139-004	10.0000	6.0000	13.0000	0.0010
139-016	14.0000	11.0000	16.0000	0.0000
139-011	14.0000	12.0000	16.0000	0.0000
139-001	14.0000	8.5000	16.0000	0.0005
139-003	16.0000	13.0000	16.0000	0.0000
139-012	17.0000	16.0000	17.0000	0.0000

Table S5: 50,000-draw integrated-domain weight sensitivity.

Component	Median rank	Low	High	Top-3 frequency
139-009	1.0000	1.0000	9.0000	0.7740
139-018	3.0000	2.0000	11.0000	0.6794
139-015	4.0000	2.0000	7.0000	0.4404
139-013	5.0000	4.0000	11.0000	0.0000
139-006	5.0000	2.0000	8.0000	0.2189
139-007	6.0000	5.0000	9.0000	0.0000
139-008	7.0000	2.0000	9.0000	0.2542
139-017	8.0000	1.0000	15.0000	0.3054
139-005	9.0000	3.0000	11.0000	0.1017
139-014	10.0000	7.0000	13.0000	0.0000
139-010	10.0000	6.0000	10.0000	0.0000
139-004	12.0000	1.0000	15.0000	0.2260
139-016	13.0000	4.0000	16.0000	0.0000
139-001	14.0000	12.0000	15.0000	0.0000
139-012	15.0000	12.0000	17.0000	0.0000
139-011	16.0000	13.0000	17.0000	0.0000
139-003	16.0000	14.0000	17.0000	0.0000

5 Spatial benchmark protocol and complete diagnostics

The input SHA-256 is 3188160a144c5020e7d21bb50a195e07a686d941134ade35a57232fe41ddc1b1. The outer development/buffer/test counts are 12,094/4,467/4,188. The minimum observed inter-sample distance is 2,040 m; the nominal grid-boundary separation is 2,010 m. The observed distance is used for reporting.

Inner validation covers block rows 0–2, 3–6 and 7–9, with adjacent block rows excluded from training. One fold has no water-class training observations. The six-class scoring denominator remains fixed. All model families receive three configurations; there is no test-driven hyperparameter search. Random forest and extra trees use 300 trees; histogram boosting 250 iterations; XGBoost and CatBoost 300 iterations; MLP a (64,32) hidden layout and 400 iterations. Complete parameters and per-fold scores are in the executable notebook and JSON search record. Early stopping is disabled. All families share the same split and selection protocol.

Table S6: Paired 2,000-block-bootstrap macro-F1 differences: primary random forest minus comparator. Intervals are unadjusted exploratory comparisons.

Comparator	F1 difference	CI low	CI high
Logistic	0.0145	0.0028	0.0283
ExtraTrees	-0.0049	-0.0099	0.0014
HistGradientBoosting	-0.0248	-0.0663	-0.0117
MLP	-0.0281	-0.1063	0.0504
XGBoost	-0.0054	-0.0587	0.0094
CatBoost	0.0290	-0.0051	0.0990

Table S7: Development-only temperature calibration evaluated on the outer stripe.

Variant	Temperature	Log loss	Brier	ECE
raw	1.0000	0.4977	0.2689	0.0143
OOF-temperature	1.6583	0.5735	0.2999	0.1357

Table S8: Fixed-capacity diagnostic controls; not alternative model-selection candidates.

Control	Macro-F1	Agreement	Log loss	ECE
bands-only	0.8245	0.7954	0.5619	0.0513
indices-only	0.6706	0.6566	0.8156	0.0337
location-only	0.1404	0.2290	1.9122	0.1467
shuffled-development-labels	0.0699	0.1538	1.7117	0.0627
class-prior	0.0495	0.1743	1.7062	0.0410

Table S9: Three-seed sensitivity of the development-selected random forest.

Seed	Macro-F1	Agreement	Log loss	ECE
311.0000	0.8214	0.8128	0.4977	0.0143
733.0000	0.8214	0.8128	0.5002	0.0142
2026.0000	0.8205	0.8116	0.4987	0.0133

6 Complete sensitivity experiments

The main experimental robustness figure presents selected components and contrasts. The tables below report every component in the joint simulation and structural scenarios, together with complete baseline and ablation comparisons. Spatial states and decision settings remain nested; frequencies are conditional simulation summaries.

Table S10: All components under 500 spatial states crossed with ten decision settings.

Component	Median rank	Low	High	Top-3	Top-5
139-009	2.000	1.000	9.000	0.732	0.861
139-017	3.000	1.000	13.000	0.526	0.624
139-018	4.000	1.000	11.000	0.463	0.728
139-015	4.000	1.000	14.000	0.410	0.751
139-006	7.000	1.000	12.000	0.158	0.326
139-007	7.000	1.000	17.000	0.118	0.373
139-013	7.000	3.000	17.000	0.062	0.301
139-014	7.000	2.000	15.000	0.046	0.213
139-010	9.000	1.000	14.000	0.153	0.250
139-008	9.000	2.000	16.000	0.059	0.116
139-004	10.000	1.000	15.000	0.164	0.242
139-005	11.000	2.000	15.000	0.063	0.118
139-001	12.000	3.000	17.000	0.032	0.052
139-011	13.000	10.000	17.000	0.000	0.000
139-016	15.000	4.000	17.000	0.014	0.042
139-003	15.000	10.000	17.000	0.001	0.002
139-012	17.000	12.000	17.000	0.000	0.000

Table S11: All components across 27 support, weight and terrain-formulation scenarios.

Component	Median rank	Low	High	Top-3	Top-5
139-009	2.000	1	8	0.630	0.852
139-018	2.000	1	7	0.741	0.852
139-015	3.000	1	11	0.667	0.889
139-007	5.000	1	13	0.259	0.556
139-013	5.000	2	15	0.074	0.630
139-006	6.000	1	10	0.222	0.407
139-008	8.000	3	14	0.074	0.111
139-014	8.000	3	13	0.074	0.074

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Table S11: All components across 27 support, weight and terrain-formulation scenarios.

Component	Median rank	Low	High	Top-3	Top-5
139-017	8.000	3	14	0.074	0.185
139-005	10.000	3	15	0.037	0.111
139-010	10.000	3	14	0.037	0.148
139-001	12.000	10	15	0.000	0.000
139-004	12.000	1	15	0.111	0.185
139-016	14.000	9	17	0.000	0.000
139-012	15.000	10	17	0.000	0.000
139-003	16.000	6	17	0.000	0.000
139-011	16.000	12	17	0.000	0.000

Table S12: Complete factor and domain ablation comparisons.

Scenario	Spearman rho	Top-5 overlap	Max shift
Full hierarchy	1.000	5	0.000
Landscape only	0.325	1	11.000
Terrain only	0.830	4	7.000
Without NDVI	0.991	5	1.000
Without NDBI	0.990	4	2.000
Without MNDWI	-0.280	0	13.000
Without slope	0.611	3	8.000
Without wetness	0.892	4	4.000
Without drainage	0.454	2	8.500

7 Exact decision-weight intervals

All pairwise crossings of the fixed E2024 linear scores were enumerated independently at each radius. Interval-interior ranks were checked at two additional points per interval (440 checks across 220 intervals). Integrating top-three membership over these intervals gives exact acceptability under a uniform weight distribution. The 500 m results differed from the recorded 50,000-draw uniform-weight experiment by at most 0.002382. These are deterministic calculations from existing summaries; they do not introduce new field observations. Boundary ties follow component-inventory order. The full machine-readable rank intervals and source checksum accompany the analysis.

Table S13: Exact first-priority weight intervals for fixed E2024 summaries and equal-factor terrain. Intervals refer to their interiors; a boundary can tie neighbouring leaders.

Radius (m)	Lower w	Upper w	First-priority component
250	0.0000	0.2941	Jaulian
250	0.2941	0.7306	Giri Mosque / tombs
250	0.7306	1.0000	Sirkap
500	0.0000	0.1047	Bhallar
500	0.1047	0.7692	Giri complex
500	0.7692	0.7857	Kalawan
500	0.7857	1.0000	Sirkap
1000	0.0000	0.7143	Jaulian
1000	0.7143	0.7273	Khader Mohra
1000	0.7273	1.0000	Dharmarajika

8 Additional environmental diagnostics

Source manifests provide retained scene identifiers, acquisition windows and preprocessing specifications. The following figures supply acquisition-specific weather, terrain and indicator-dependence details supporting the main analysis.

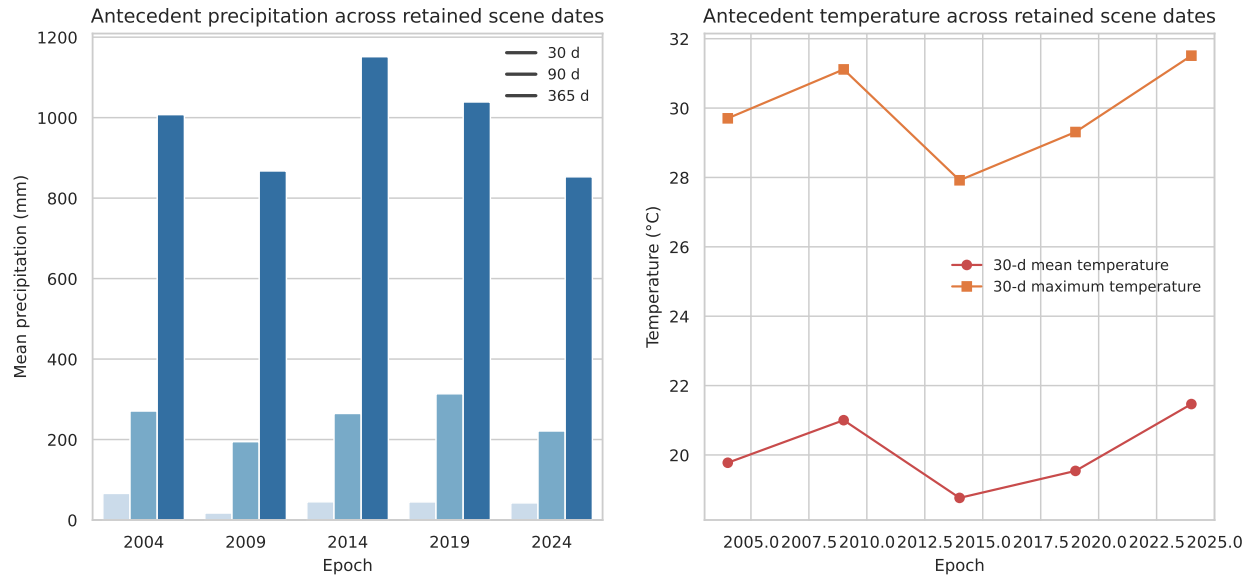


Figure S1: Weather matched to the retained Landsat acquisition windows; contextual rather than causal.

Terrain and hydrological susceptibility context

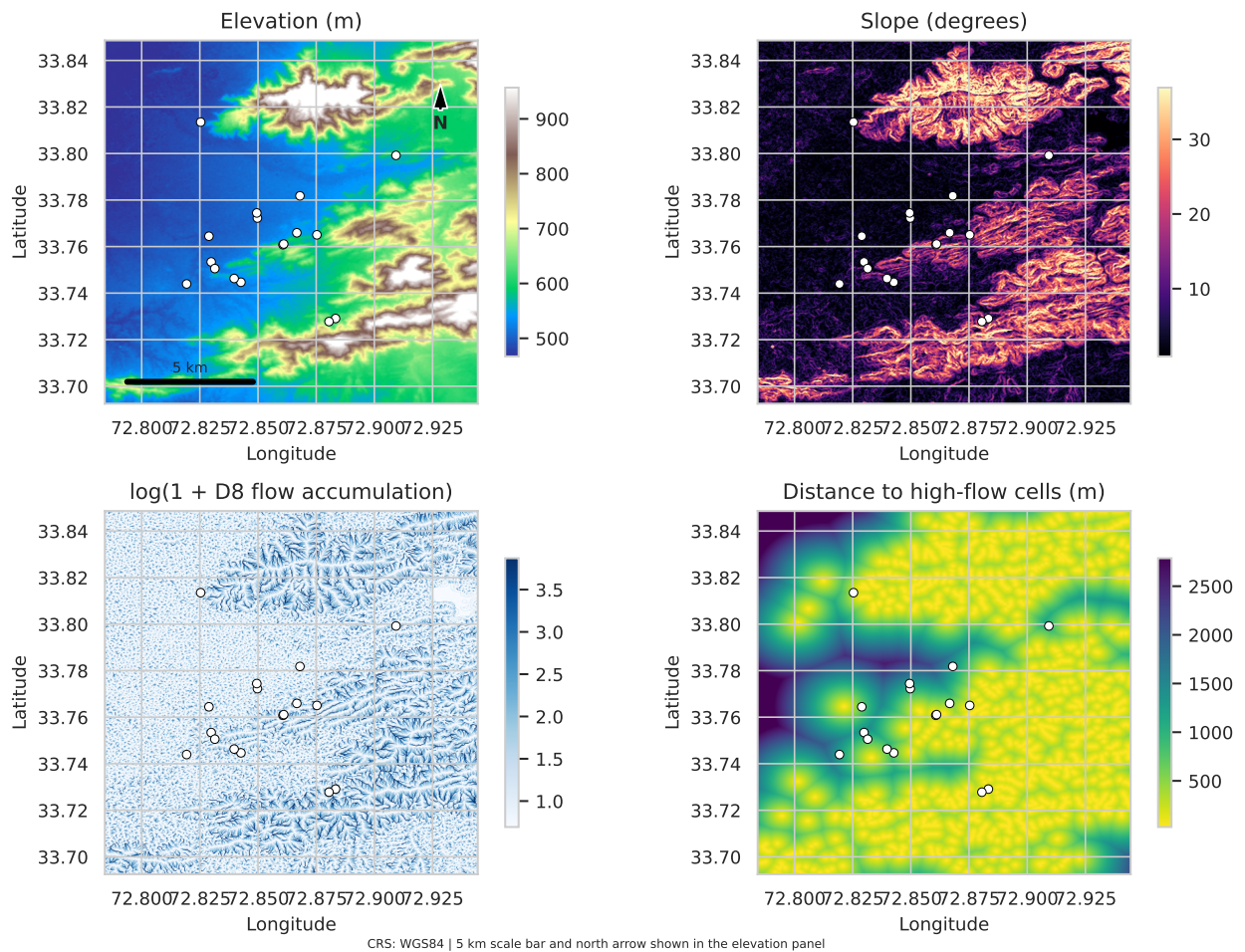


Figure S2: Elevation-derived terrain and drainage proxies. These do not constitute a hydraulic model.

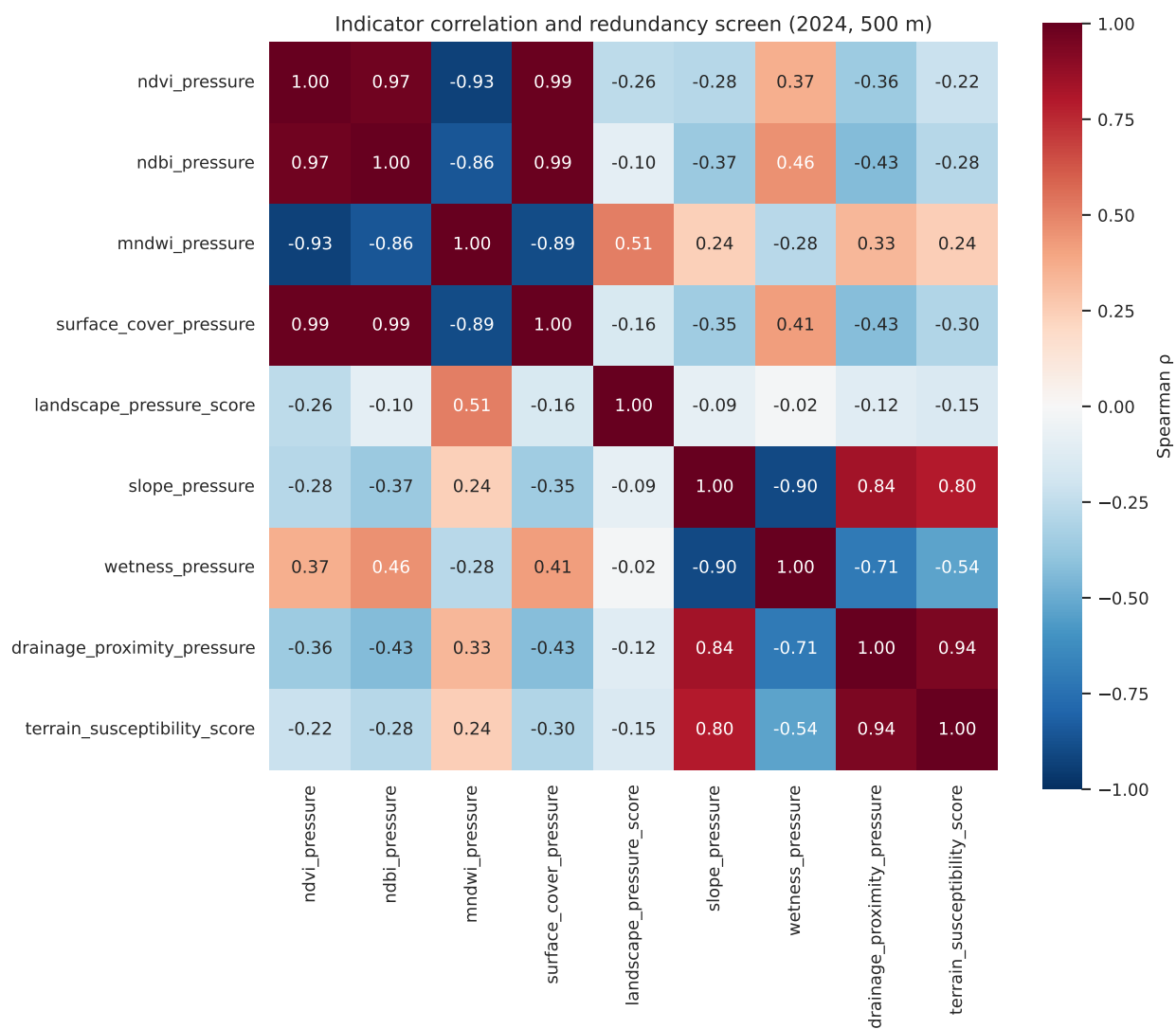


Figure S3: Dependence among component-level indicator ranks.